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10 / 10 submitted

Q1. Sum the given number upto single digit

Score: 5.00 TC: 1/1 passed

CODE

```
n = int(input().strip())
while n ≥ 10:
    s = 0
    while n > 0:
        s += n % 10
        n //= 10
    n = s

print(n)
```

INPUT

89789

OUTPUT

5

Q2. Prime Number

Score: 5.00 TC: 1/1 passed

CODE

```
def is_prime(n):
    if n < 2:
        return False
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
    return True

n = int(input().strip())

x = n + 1
while True:
    if is_prime(x):
        print(x)
        break
    x += 1
```

INPUT

13

OUTPUT

17

Q3. Swap the first and last digits

Score: 5.00 TC: 1/1 passed

CODE

```
n = input().strip()
if len(n) == 1:
    print(n)
else:
    first = n[0]
    last = n[-1]
    middle = n[1:-1]
    print(last + middle + first)
```

INPUT

23546

OUTPUT

63542

Q4. Odd Sequence first and Even Sequence next

Score: 5.00 TC: 1/1 passed

CODE

```
n = input().strip()
odds = []
evens = []

for ch in n:
    d = int(ch)
    if d % 2 != 0:
        odds.append(ch)
    else:
        evens.append(ch)

print("".join(odds) + "".join(evens))
```

INPUT

658134

OUTPUT

513684

Q5. Step Number

Score: 5.00 TC: 2/2 passed

CODE

```
n = input().strip()

# remove leading sign if any (optional)
if n[0] == '-':
    n = n[1:]

# if n has 0 or 1 digits, it's always a step number
if len(n) <= 1:
    print("Yes")
else:
    is_step = True
    for i in range(len(n) - 1):
        if abs(int(n[i]) - int(n[i + 1])) != 1:
            is_step = False
            break

    print("Yes" if is_step else "No")
```

INPUT

54567

OUTPUT

Yes

Q6. Reverse only the even numbers

Score: 5.00 TC: 1/1 passed

CODE

```
s = input().strip()

evens = [ch for ch in s if int(ch) % 2 == 0]
evens.reverse()

j = 0
res = []
for ch in s:
    if int(ch) % 2 == 0:
        res.append(evens[j])
        j += 1
    else:
        res.append(ch)

print("".join(res))
```

INPUT

325649

OUTPUT

345629

Q7. Encoding

Score: 5.00 TC: 1/1 passed

CODE

```
num = input().strip()

count = 1
out = []

for i in range(1, len(num) + 1):
    if i < len(num) and num[i] == num[i - 1]:
        count += 1
    else:
        out.append(str(num[i - 1]))
        out.append(str(count))
        count = 1

print(''.join(out))
```

INPUT

55553322255

OUTPUT

54322352

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Score: 0.00 TC: 0/1 passed

CODE

```
n = input().strip()
digits = list(n)

odd = [d for d in digits if int(d) % 2 != 0]
odd.reverse()

j = 0
for i in range(len(digits)):
    if int(digits[i]) % 2 != 0:
        digits[i] = odd[j]
        j += 1

print(''.join(digits))
```

INPUT

816253

OUTPUT

836251

Q9. Convert every number into the next sequence number.

Score: 5.00 TC: 1/1 passed

CODE

```
n = input().strip()
ans = ""
for ch in n:
    d = int(ch)
    ans += str((d + 1) % 10)
print(ans)
```

INPUT

89781

OUTPUT

90892

Q10. Print the total factor count of a given number

Score: 5.00 TC: 1/1 passed

CODE

```
n = int(input())

count = 0
for i in range(1, n + 1):
    if n % i == 0:
        count += 1

print(count)
```

INPUT

12

OUTPUT

6